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1. An acute angled triangle ABC has orthocentre H . The circle passing through H with centre the midpoint of BC intersects BC at points A_1 and A_2 . Similarly define points B_1, B_2, C_1, C_2 . Show that the six points $A_1, A_2, B_1, B_2, C_1, C_2$ lie on a circle.
2. (a) Prove that $x^2/(x-1)^2 + y^2/(y-1)^2 + z^2/(z-1)^2 \geq 1$ for all real numbers x, y, z not equal to 1 satisfying $xyz = 1$.
(b) Prove that equality holds for infinitely many rational triples (x, y, z) .
3. Prove that there exist infinitely many positive integers n such that $n^2 + 1$ has a prime divisor greater than $2n + (2n)^{1/2}$.
4. Find all functions f from $(0, +)$ to $(0, +)$ such that $(f(w)^2 + f(x)^2)/(f(y^2) + f(z^2)) = (w^2 + x^2)/(y^2 + z^2)$ for all positive real numbers w, x, y, z satisfying $wx = yz$.
5. Let n and k be positive integers with $k \geq n$ and $k - n$ even. There are $2n$ lamps, initially all off. At each step one lamp is switched.
Let N be the number of sequences of k steps resulting in lamps 1 to n on and lamps $n + 1$ to $2n$ off.
Let M be the number of such sequences where lamps $n + 1$ to $2n$ are never switched on. Find N/M .
6. Let $ABCD$ be a convex quadrilateral with $|BA|$ not equal to $|BC|$. Let the incircles of triangles ABC and ADC be considered. Suppose a circle is tangent to ray BA beyond A and ray BC beyond C , and also tangent to lines AD and CD . Prove that the common external tangents of the two incircles intersect on this circle.